

Project title: SplitsFromVideo

Team Members:

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Faculty Advisor: Eraldo Ribeiro (eribeiro@fit.edu)

Client: Speedrunning community (speedrun.com)

Task	Completion %	Mateusz	Carter	To Do
1. Compare and select Technical Tools	100%	50%	50%	none
2. "hello world" demos	100%	50%	50%	Demo of the PARTS algorithm
3. Resolve Technical Challenges	100%	50%	50%	none
4. Compare and select Collaboration Tools	100%	50%	50%	none
5. Requirement Document	100%	30%	70%	none
6. Design Document	100%	70%	30%	none
7. Test Plan	100%	50%	50%	none

Discussion of tasks:

- Task 1: We compared tools and decided that the selected tools from the project plan should be sufficient.
- Task 2: We implemented two test Python programs. One opens both a video and audio stream using yt-dlp, and the other uses cap-from-youtube to download frame 300 of the specified video from the link.

- Task 3: We discussed some of the technical challenges and are confident we can overcome them.
- Task 4: This task was very self explanatory. We chose Discord as our main form of communication, and Google Docs/Google Calendar to stay organized.
- Task 5: The requirements document is completely finished and outlines the requirements of the program very well.
- Task 6: The design document is also finished and outlines the design well.
- Task 7: The testing document is also finished and outlines the testing well.

Discussion of contribution:

- Mateusz Doda: Design Document, Testing Plan
- Carter Tabin: Hello world demo, requirements document, testing plan

Plan for the next milestone:

Task	Mateusz	Carter
Implement, test, and demo navigating and obtaining specific frames from a YouTube video.	Get more comfortable with cap-from-youtube and its arguments.	Get more comfortable with cap-from-youtube and its arguments.
Implement, test, and demo algorithm for finding the beginning of splits.	Figure out how to manage frame advancing efficiently. And implement the basis of the algorithm	Figure out how to manage frame advancing efficiently. And implement the basis of the algorithm

Discussion (at least a paragraph) of each planned task for the next Milestone

- Task 1: Implement, test, and demo navigating and obtaining specific frames from a YouTube video.

For this milestone, we plan to have a fully fleshed out system for navigating to specific frames of the video. While we were able to implement this as a demo for milestone 1, we plan to improve upon it for milestone 2. For this implementation, our plan is to come to a full understanding of how cap-from-youtube and yt-dlp function in terms of obtaining frames, as well as figure out how to only get a specific section of the frame rather than the whole image, since we only need the section containing the timer. Once we are able to do this, it will be time to start implementing computer vision and OCR on the images and extracting timer data.

- Task 2: Implement, test, and demo algorithm for finding the beginning of splits.

For this milestone, we plan to implement the actual algorithm for finding the beginning of splits. This implementation will be more basic than the final design. For this milestone, instead of using OCR to extract the timer data, we will pop up the frame on screen and prompt the user for the timer data manually. While this does not fix the problem the algorithm is trying to solve entirely, it allows a good baseline for testing the algorithm in a way that doesn't depend on fully functional and dependable OCR implementation.

Date(s) of meeting(s) with Client during the current milestone: N/A

Client feedback on the current milestone: N/A

We chose not to meet with any clients during this milestone. The main "clients" we used for the project proposal were speedrunners from Discord. We chose not to contact these speedrunners for this milestone due to how clear and simple the requirements and design documents were. When we get to more nuanced or user-centric requirements, we may talk with the previous clients again.

Date(s) of meeting(s) with Faculty Advisor during the current milestone: N/A for current milestone. Discussion of project before class Intro to CV class.